

FlashAttention-v2: Faster Attention with Long-sequence Hallucination Study

Abstract

We propose FlashAttention-v2, an Algorithm that improves upon Transformer baseline. Evaluated on HumanEval and ImageNet-1k, it improves BLEU-4 by 2 points and reduces Latency. It requires Cross-Entropy with Label Smoothing and is vulnerable to Quadratic Memory Footprint. See Figure 2 and Table 1 for main results.

1. Introduction

Long-sequence Hallucination is a key Problem. Prior work fails on long contexts. Our contribution: faster attention with lower memory. Conclusion preview: gains hold on benchmarks.

2. Methodology

Model Formulation: we modify attention with tiling. Loss is Cross-Entropy with Label Smoothing. The model requires careful IO scheduling. See Sec 3.2 for settings.

3. Experiments

Evaluation: on HumanEval BLEU-4 improves 2 points. Ablation in Table 1 shows tiling matters. Figure 2 shows latency vs length. Compared to Transformer baseline, our method wins. Limitation: Quadratic Memory Footprint remains for extreme lengths.

4. Conclusion

FlashAttention-v2 improves upon Transformer, evaluated on HumanEval and ImageNet-1k. Future work addresses Quadratic Memory Footprint.